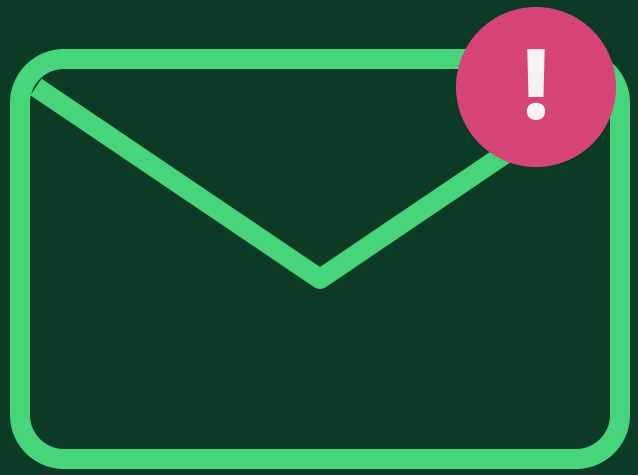
SpamShield

99% → 20%

How 30 innocent words fool a spam filter



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INTRODUCTION

Spam filters look almost perfect. Are they?

Spam carries scams, phishing and malware.

Filters **learn** from labeled emails: spam vs. normal email (“ham”).

But **spammers adapt** — they change their emails to slip through.

RQ1 · Which model is best?

Naive Bayes vs. Logistic Regression vs. a neural network

RQ2 · Can we fool them?

Change letters, add spaces, add innocent words

RQ3 · Can we fix it?

Two defenses, tested fairly

DATA

165,059 real emails → 122,850 unique

TREC 2005
92,189

CEAS 2008
39,154

Enron-Spam
33,716

→

Clean + remove duplicates
42,209 removed

→

122,850
50% spam · 50% ham

BEFORE

```
RE: You have WON $1,000!!! <p>Click <a
href="http://prize.example.com">HERE</a>
to claim your prize before 12/31!</p>
```

→

AFTER

```
re you have won num num
click here to claim your
prize before num num
```

METHODOLOGY

From words to a spam score

Clean text
subject + body

→

TF-IDF
50,000 words & word pairs → numbers

→

Model
3 kinds, below

→

P(spam)
> threshold → spam

Naive Bayes

Word probabilities per class. Simple, very fast **baseline**.

Logistic Regression

One **weight per word**, added up → sigmoid. Easy to explain.

Neural network (MLP)

50,000 inputs → **64 ReLU** → **1 sigmoid**

Adam, batch 256, **early stopping** at epoch 4

RESULTS · RQ1

All three models catch 95–99% of spam

Naive Bayes

F1 score

0.967

Spam caught: **94.7%**

False alarms: **1.3%**

Training time: **0.06 s**

Logistic Regr.

F1 score

0.985

Spam caught: **98.7%**

False alarms: **1.9%**

Training time: **1.2 s**

40× faster than the neural net!

Neural net (MLP)

F1 score

0.987

Spam caught: **98.9%**

False alarms: **1.5%**

Training time: **46 s**

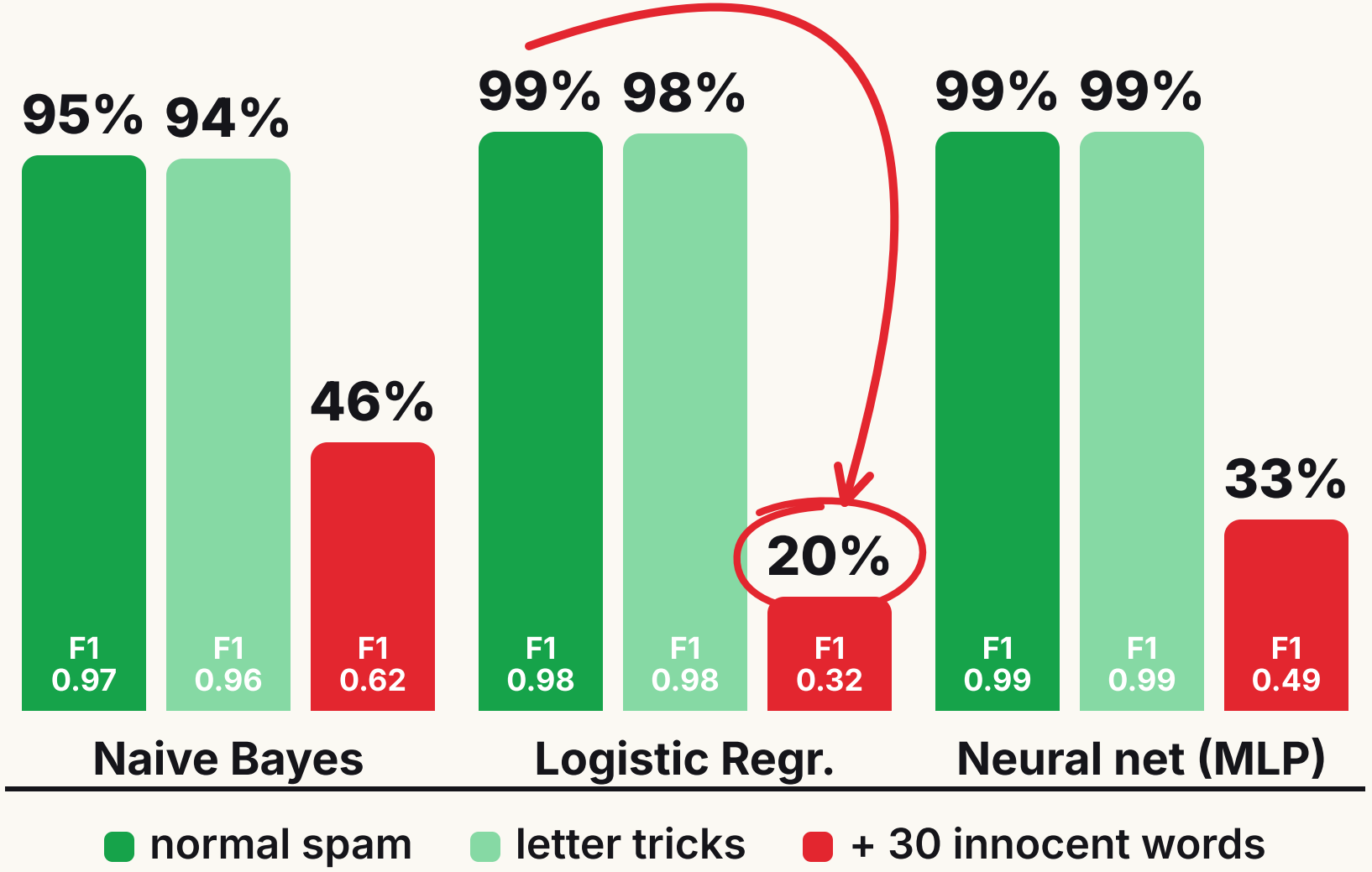
EXPERIMENT 2 · RQ2

We disguised the spam in the test set

Original spam	`buy cheap viagra and replica watches save money fast`
Leet: letters → digits	`buy cheap v1agra and repl1ca w4tch3s s4ve mon3y f4st`
Spacing	`buy cheap v i a g r a and r e p l i c a w a t c h e s s a v e m o n e y f a s t`
+ 30 office words	`buy cheap viagra and replica watches save money fast meeting project thanks schedule report team …`
+ 30 typical ham words	`buy cheap viagra and replica watches save money fast wrote forwarded kaminski python houston mike …`

EXPERIMENT 2 · RESULT

Letter tricks fail. Recall and F1 dropped.



The most accurate models are the **EASIEST** *to fool!*

EXPERIMENT 2 · WHY?

The model just weighs words — like a scale

Spam words (+)

- `viagra` +6.0
- `http` +8.5
- `your` +8.5
- `remove` +6.7



Ham words (–)

- `enron` -15.4
- `thanks` -13.9
- `wrote` -13.1
- `vince` -9.3

EXPERIMENT 3 · FIX 1

Fix 1: don't throw the whole email into one bag

BEFORE

54 spam words
viagra cialis …
+ 60 ham words
wrote thanks python …

→



→

whole email:
0.06
HAM
✗ fooled

FIX 1

54 spam words
viagra cialis …
+ 60 ham words
wrote thanks python …

→

piece 1 · 50 words
viagra click
free …
1.00

piece 2 · 64 words
wrote mike
python …
0.00

→

highest piece:
1.00
SPAM
✓ caught

EXPERIMENT 3 · FIX 2

Fix 2: teach the model with disguised spam

1 · We added disguised spam to the training data

spam email

`viagra` `click` `free` …

→ SPAM

normal email

`thanks` `meeting` `draft` …

→ HAM

NEW: spam + random ham words

`viagra` `click` + `tony` `rant` …

→ still SPAM

2 · Result: it learned the words, not the trick

Words it saw in training → weaker ✓

word weight: before → after

`tony`	-6.2 → -4.3
`rant`	-5.7 → -4.0
`pdf`	-5.2 → -3.5

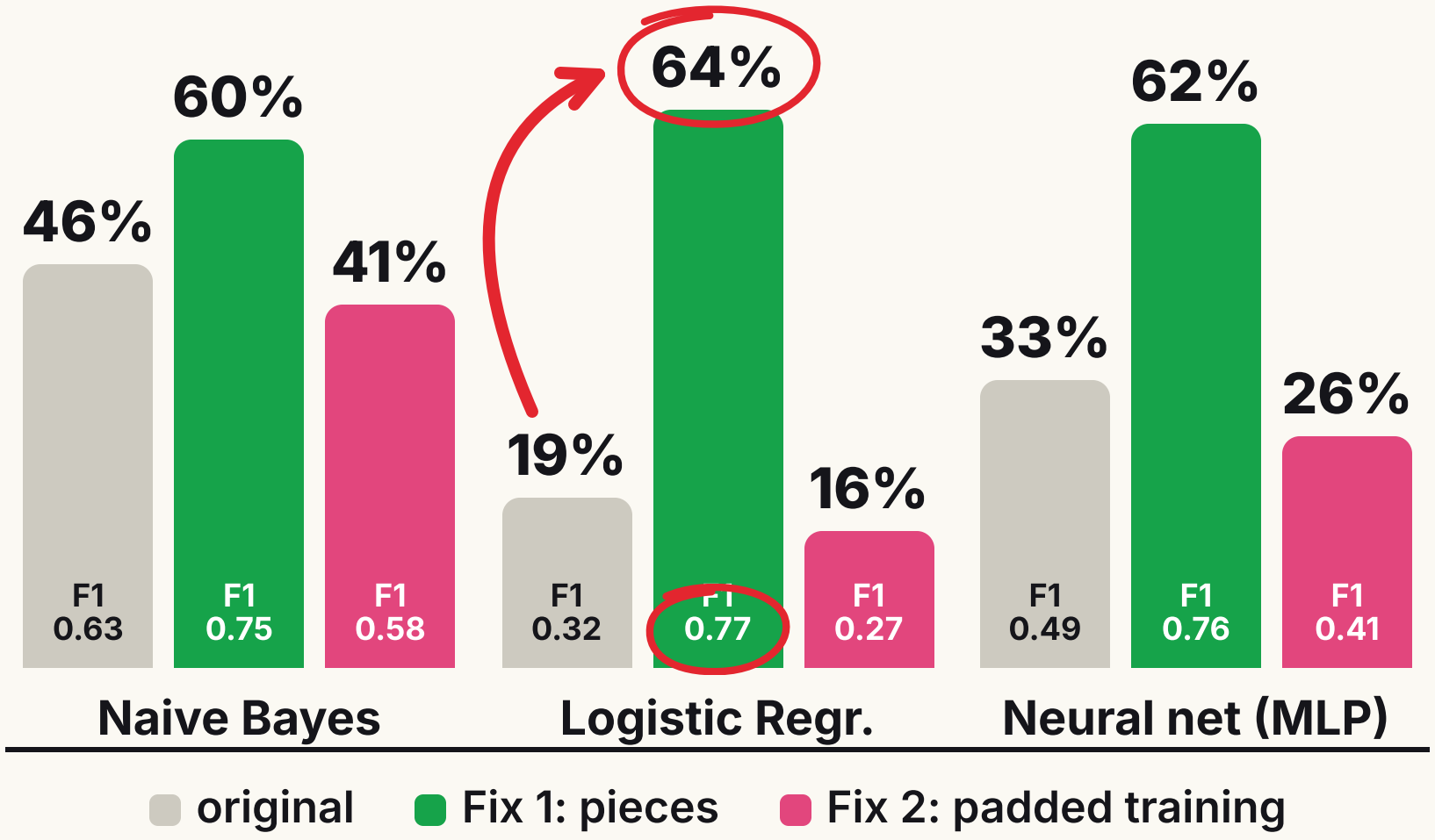
Attacker's words → still strong ✗

word weight: before → after

`thanks`	-13.9 → -15.8
`python`	-6.4 → -7.3
`mike`	-4.6 → -5.6

EXPERIMENT 3 · RESULT

Fix 1 works against any words. Fix 2 doesn't.



3.3×

more spam caught by Logistic Regression!

DISCUSSION & CONCLUSION

What we learned

1. Logistic Regression is the best everyday choice: almost as accurate as the neural net, **40× faster**, explainable.

2. Word-based filters are easy to fool by **adding innocent words** — even the neural net.

3. Retraining protects only against words the model has **already seen**.

4. Scoring in **pieces** protects against any added words — at a cost.

Limits

Old data (2005–2008). An attacker could hide words *inside* every piece.

Next steps

Test on newer spam (train 2005 → test 2008), tune piece size, try DistilBERT, add sender & link signals.

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Full reference list in the research paper.